

Evaluating the R-Super MedFormer model on PanTS

with the SegEvalKit evaluation library

901 test CT scans • 17 structures • organ, tumour and patient-level metrics • evaluated 2026-09-30 with SegEvalKit 0.1.0

At a glance

What we tested

Model	MedFormer , the segmentation model of the R-Super project (Report Supervision, Johns Hopkins University). This is the only R-Super model we ran. It is the public checkpoint <code>AbdomenAtlas/MedFormerPanTS</code> , which the authors describe as their “segmentation” baseline, trained on the PanTS public dataset; the report-supervised R-Super models were not evaluated here.
Code	Official R-Super repository (<code>MrGiovanni/R-Super</code> , commit <code>fa8fdce</code> , 2026-09-24), official inference script <code>predict_abdomenatlas.py</code> , with one disclosed bug fix (Section 1.2).
Dataset	PanTS (Pancreatic Tumor Segmentation dataset), public test split : 901 contrast and non-contrast abdominal CTs with expert per-structure masks and patient metadata.
Comparator	TotalSegmentator v2 (total task), a widely used general model, as a reference point only.
Evaluation	SegEvalKit 0.1.0: every metric on every case, then test-set, patient-level, paired and cohort statistics. Every number in this report is generated from the evaluation output files.

.864

pancreas Dice

.971

liver Dice

2.9 mm

pancreas HD95 (median)

0.905

tumour yes/no AUC

0.81

tumours found

0.49

false tumours / scan

Five things the numbers say

- Strong organ segmentation.** MedFormer’s mean Dice is at least 0.85 on 11 of its 16 organs and vessels (pancreas .864, median .910). On the ten organs it shares with TotalSegmentator it has the higher Dice in 79% to 98% of the patients.
- A few cases drive the average boundary error.** On the spleen, 79 of 901 cases with an HD95 above 50 mm make up 94% of the mean HD95 (median 1.5 mm, mean 31.7 mm). 29 of them are patients with no reference spleen mask in which MedFormer still predicts a spleen; the other 50 are good segmentations (median Dice .957) with some predicted voxels far from the organ.
- How you rank decides who wins.** On the spleen and the kidney (left), ranking by mean Dice puts TotalSegmentator first, while counting in how many patients each model is better puts MedFormer first.
- Tumours: sensitive, not precise.** At the patient level MedFormer separates patients with and without a pancreatic tumour with an AUC of 0.905. It finds 0.81 of all 161 reference tumours, but only 0.24 of its predicted tumours match a real one (0.49 false tumours per scan), and tumours smaller than 0.5 mL are almost always missed (4% of 25).
- Performance depends on the scan.** Pancreas Dice is .884 on venous-phase and .806 on non-contrast scans, and .850 on Siemens against .719 on GE scanners; 25 of 63 cohort tests are significant after correction.

1 What we tested, and how

1.1 The model: MedFormer from R-Super

R-Super (Report Supervision) is a method from Johns Hopkins University that trains tumour segmentation models with radiology reports in addition to masks (Bassi *et al.*, MICCAI 2025). Its repository releases a MedFormer model trained with masks on PanTS, the checkpoint `AbdomenAtlas/MedFormerPanTS`, which its model card calls the public “segmentation” baseline and the starting point for report supervision. We evaluate *this* checkpoint (`pants_pancreas_release/fold_0_latest.pth`), the only R-Super model we ran. It predicts 26 classes, including the pancreas, its head, body and tail, the pancreatic lesion and the surrounding organs and vessels, and it writes lesion probabilities.

We ran the official script `predict_abdomenatlas.py` on NVIDIA A100 GPUs of the MBZUAI DANA cluster, as SLURM batch jobs (`benchmarks/pants/run_medformer.sbatch`), with the options

```
--organ_mask_on_lesion --save_probabilities_lesions --organs_with_tumor pancreas
```

1.2 One fix to the official code

What went wrong. On 25 of the 901 scans the official script finished the network prediction but failed when writing the result. The cause is in `pad_to_training_size`: PyTorch’s `F.pad` lists the *last* axis first, so a scan with fewer than 128 voxels along one axis (after resampling to 1 mm) is padded along the wrong axis while the offsets are recorded for the right one. Removing the padding then cuts the wrong axis and the output no longer matches the CT (23 scans short along *z*, 2 narrow along *x*).

What we did. We ran a copy of the script with the two padding lines swapped (`benchmarks/pants/patch_medformer.py`); the official file is unchanged. The fixed lines only run for such small scans, so the other 876 predictions are identical with and without the fix.

1.3 The data: PanTS test split

PanTS is a public CT dataset of the pancreas and pancreatic tumours. Its test split has 901 CT scans, of which 151 contain a pancreatic tumour and 750 do not. Each scan comes with one mask file per structure, and a metadata table gives the CT phase, scanner, site country, slice thickness, sex and age.

We evaluate 17 structures: the pancreas, the pancreatic lesion, liver, spleen, both kidneys, stomach, gallbladder, duodenum, colon, both adrenal glands, aorta, inferior vena cava, superior mesenteric artery, veins and common bile duct.

Two dataset details that change the numbers.

- **Pancreas = pancreas + tumour.** In PanTS the tumour mask is not always inside the pancreas mask, so the pancreas is scored as the union of the two, on both the reference and the prediction.
- **Unreliable label headers.** `segevalkit audit` finds that in 227 of the 901 scans at least one reference mask (381 files, 218 of them aortas) stores an orientation or origin that disagrees with its CT, although its voxels line up with the CT and all of them have the CT’s shape. Aligning masks by their headers would move correct predictions out of place, so every mask is compared voxel by voxel with the CT grid.

1.4 The evaluation protocol

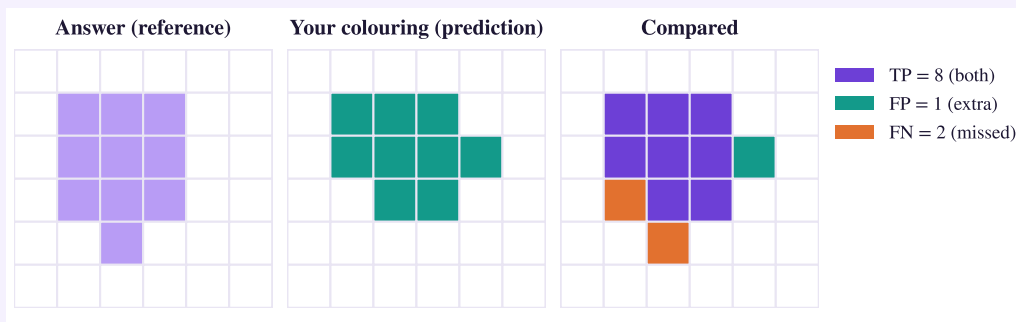
- **Organs:** Dice, IoU, NSD at 2 mm, HD95, ASSD, MASD, precision, recall and volume errors. Tubular structures (aorta, vena cava, veins, arteries, bile duct) also get cDice and Betti errors.
- **Pancreatic lesion:** the organ metrics plus lesion-level detection (tumours are connected blobs, 26-connectivity, blobs under 10 voxels ignored), panoptic quality, lesion counts and probability calibration.
- **Empty masks:** if both reference and prediction are empty the case is perfect (Dice 1, HD95 0); if only one is empty, overlap scores are 0 and distances are the image diagonal. A missing prediction counts as empty.
- **Test set:** per-case means with 95% bootstrap intervals, pooled (voxel-summed) values, patient-level tumour detection, paired tests against TotalSegmentator, two ranking schemes and cohort tests on the metadata.

PanTS does not publish its evaluation script, NSD tolerance or detection rule, so these numbers follow the documented SegEvalKit protocol and are not directly comparable with the PanTS leaderboard.

2 How each metric works, explained simply

The picture game

A CT scan is a big 3D picture made of tiny cubes called *voxels*. Doctors coloured the cubes that belong to, say, the pancreas: that is the **answer key** (the *reference*, G). The model coloured its own guess: the **prediction** P . Comparing the two gives three counts: **TP** (coloured by both), **FP** (coloured by the model only, extra) and **FN** (in the answer, but missed). Every overlap metric is built from these.



The examples below use this picture: the answer has 10 squares, the colouring has 9, and 8 are shared, so $TP = 8$, $FP = 1$, $FN = 2$.

2.1 Overlap: how much of the shape is right?

Dice similarity coefficient (DSC)

$$DSC = \frac{2TP}{2TP + FP + FN}$$

Like you're 10: Count the shared squares twice and divide by all the squares coloured in both pictures. It is the most common segmentation score.

Good values: 0 (no overlap) to 1 (perfect). Higher is better.

Tiny example: $2 \cdot 8 / (2 \cdot 8 + 1 + 2) = 16/19 = 0.84$.

Intersection over union (IoU)

$$IoU = \frac{TP}{TP + FP + FN}$$

Like you're 10: Shared squares divided by all squares that anyone coloured. It is always a bit lower than Dice and tells the same story.

Good values: 0 to 1. Higher is better.

Tiny example: $8 / (8 + 1 + 2) = 8/11 = 0.73$.

Precision and recall

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Like you're 10: Precision: of the squares you coloured, how many were right? Recall: of the squares you should have coloured, how many did you colour? A model that colours too much has low precision; one that colours too little has low recall.

Good values: 0 to 1. Higher is better.

Tiny example: Precision $8/9 = 0.89$; recall $8/10 = 0.80$.

Relative volume difference (RVD)

$$\text{RVD} = \frac{|P| - |G|}{|G|}$$

Like you're 10: Is the amount right? Compare the number of coloured squares with the answer, as a fraction. It ignores *where* the squares are.

Good values: 0 is perfect; -0.2 means 20% too small, $+0.2$ means 20% too big.

Tiny example: $(9 - 10)/10 = -0.10$: 10% too small.

2.2 Boundary: where is the outline wrong, in millimetres?

For these metrics we take the outline (surface) of each shape and measure, for every point of one outline, the distance in millimetres to the closest point of the other outline: $D_{P \rightarrow G}$ from the prediction to the reference and $D_{G \rightarrow P}$ back.

Normalised surface Dice (NSD at $\tau = 2$ mm)

$$\text{NSD}_\tau = \frac{N_\tau(D_{P \rightarrow G}) + N_\tau(D_{G \rightarrow P})}{|\partial P| + |\partial G|}$$

N_τ counts the distances $\leq \tau$; $|\partial P|$, $|\partial G|$ are the numbers of outline points

Like you're 10: Draw a 2 mm “safe zone” along each outline. NSD is the fraction of both outlines that lies inside the other's safe zone. Small wobbles are forgiven; real boundary mistakes are not.

Good values: 0 to 1. Higher is better.

Tiny example: Each outline has 20 points; 17 of the prediction's and 18 of the reference's are within 2 mm: $(17 + 18)/40 = 0.875$.

95th-percentile Hausdorff distance (HD95)

$$\text{HD}_{95} = \max(P_{95}(D_{P \rightarrow G}), P_{95}(D_{G \rightarrow P}))$$

Like you're 10: Line up all the outline distances from nearest to farthest. HD95 is the distance that 95% of the points are within, taken in the worse direction. It ignores the worst 5% so one stray point does not decide the score, but if more than 5% of the outline is far away, HD95 becomes large.

Good values: 0 mm is perfect. Lower is better. If one mask is empty, HD95 is the size of the whole image (hundreds of mm).

Tiny example: 100 outline points: 96 are within 1 mm and 4 are 30 mm away, so HD95 is 1 mm. With 10 points at 30 mm, HD95 is 30 mm.

Average symmetric surface distance (ASSD)

$$\text{ASSD} = \frac{\sum D_{P \rightarrow G} + \sum D_{G \rightarrow P}}{|\partial P| + |\partial G|}$$

Like you're 10: The average distance between the two outlines, using every point of both.

Good values: 0 mm is perfect. Lower is better.

Tiny example: Prediction points 0, 1, 1, 2 mm away; reference points 1, 1, 0, 0 mm: $(4 + 2)/8 = 0.75$ mm.

2.3 Tumours: were they found, and only them?

Tumours are counted as blobs (connected groups of voxels). A reference tumour is *found* if any predicted blob touches it; a predicted blob that touches no real tumour is a *false tumour*.

Lesion sensitivity, precision and F1

$$\text{Sens} = \frac{TP_{\text{les}}}{TP_{\text{les}} + FN_{\text{les}}}$$

$$\text{Prec} = \frac{TP_{\text{les}}}{TP_{\text{les}} + FP_{\text{les}}}$$

$$\text{F1} = \frac{2TP_{\text{les}}}{2TP_{\text{les}} + FP_{\text{les}} + FN_{\text{les}}}$$

Like you're 10: Hide and seek: sensitivity is how many hidden friends you found; precision is how often you were right when you shouted “found you!”; F1 combines both.

Good values: 0 to 1. Higher is better.

Tiny example: 4 real tumours, 3 found: sensitivity $3/4 = 0.75$. 5 predicted blobs, 3 on a real tumour: precision $3/5 = 0.60$. F1 = $6/(6 + 2 + 1) = 0.67$.

False-positive lesions per scan

$$\frac{\sum_{\text{scans}} FP_{\text{les}}}{\text{number of scans}}$$

Like you're 10: On average, how many false alarms does a doctor see per scan?

Good values: 0 is perfect. Lower is better.

Tiny example: 5 false blobs over 10 scans: 0.5 per scan.

Lesion-wise Dice and panoptic quality (PQ)

$$\text{LW-DSC} = \frac{\sum_i \text{DSC}_i}{N_G + FP_{\text{les}}}$$

$$\text{PQ} = \underbrace{\frac{\sum \text{IoU}}{|TP|}}_{\text{shape}} \times \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{finding}}$$

Like you're 10: Lesion-wise Dice scores every tumour separately, so a small tumour counts as much as a big one; missed and false tumours score 0. PQ multiplies “how well are the found tumours shaped” by “how well were tumours found” (a match needs IoU above 0.5).

Good values: 0 to 1. Higher is better.

Tiny example: Two tumours, Dice 0.8 and 0 (missed), plus one false blob: $0.8/3 = 0.27$. PQ with matches of IoU 0.8 and 0.6, one false and one missed: $0.7 \times 2/3 = 0.47$.

Patient level: sensitivity, specificity and AUC

$$\text{Sens} = \frac{\text{sick patients flagged}}{\text{sick patients}}$$

$$\text{Spec} = \frac{\text{healthy not flagged}}{\text{healthy patients}}$$

AUC = area under the curve of Sens vs. 1–Spec over all thresholds

Like you're 10: The doctor's question: does this patient have a tumour? The model says “yes” if its predicted tumour volume is above a threshold. The AUC is the chance that a random sick patient gets a larger predicted tumour than a random healthy one. We report sensitivity at the smallest threshold with specificity at least 0.90.

Good values: 0.5 is guessing, 1 is perfect. Higher is better.

Tiny example: 10 sick patients, 8 flagged: sensitivity 0.8. 20 healthy, 18 not flagged: specificity 0.9.

2.4 Probabilities: can the model's confidence be trusted?

MedFormer also writes a probability for every voxel of being tumour. These metrics use the voxels within 10 mm of either mask.

Expected calibration error (ECE)

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$$

Like you're 10: Like a weather forecaster: when they say “90% sure”, it should be right 9 times out of 10. Group voxels by how sure the model is ($B = 15$ groups), and average the gap between how sure it was and how often it was right.

Good values: 0 is perfectly honest. Lower is better.

Tiny example: 100 voxels at 90% confidence, 80 correct: gap 0.10. If every group looks like this, ECE is 0.10.

Brier score and area under the precision–recall curve

$$BS = \frac{1}{N} \sum_i (p_i - g_i)^2$$

$$AP = \sum_k (R_k - R_{k-1}) P_k$$

Like you're 10: Brier: the average squared miss between the probability and the truth (1 tumour, 0 not). AP: sort voxels from most to least suspicious and check whether the real tumour voxels come first; it only exists for patients with a tumour.

Good values: Brier 0 is perfect (lower is better); AP 1 is perfect (higher is better).

Tiny example: Probability 0.9 on a tumour voxel: $(0.9 - 1)^2 = 0.01$; on a healthy voxel: $0.9^2 = 0.81$; mean 0.41.

2.5 From many patients to one conclusion**Macro (per-patient) vs. pooled (per-voxel) averages**

$$\overline{DSC} = \frac{1}{n} \sum_i DSC_i$$

$$DSC_{\text{pooled}} = \frac{2 \sum_i TP_i}{2 \sum_i TP_i + \sum_i FP_i + \sum_i FN_i}$$

Like you're 10: Macro gives every patient one vote. Pooled throws all voxels of all patients into one bag, so big organs and big tumours get more say.

Good values: Both 0 to 1. When they disagree, a few patients with small structures are failing.

Tiny example: Patient A: TP 900, FP 100, FN 100 (Dice 0.90). Patient B: a 10-voxel structure missed (Dice 0). Macro 0.45; pooled $1800/2010 = 0.90$.

Bootstrap 95% confidence interval

resample the n patients with replacement, recompute the mean, repeat 1000 times; the interval holds the middle 95% of these means

Like you're 10: If we had tested a different bag of 901 patients, how much could the average change? We fake many new bags by drawing patients from our own bag (some twice, some not at all) and look at how much the average moves.

Good values: A narrow interval means a trustworthy average.

Tiny example: If the middle 95% of 1000 resampled means runs from 0.85 to 0.87, the interval is [0.85, 0.87].

Paired Wilcoxon test with Holm correction

$$W^+ = \sum_{d_i > 0} \text{rank}(|d_i|)$$

Holm:

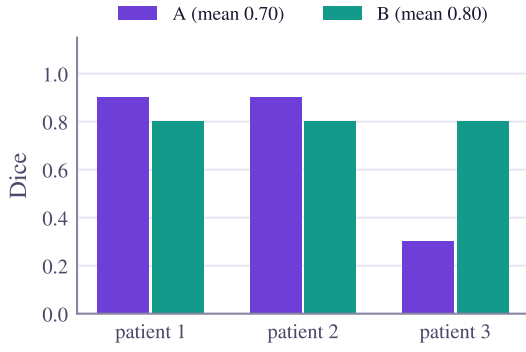
$$\tilde{p}_{(k)} = \max_{j \leq k} \min(1, (m - j + 1) p_{(j)})$$

Like you're 10: Both models take the same test on the same patients. For each patient we note who did better and by how much. If one model wins more often *and* by more, the test says it is not luck. Asking many questions at once makes some look lucky by chance, so Holm raises the bar with every extra question.

Good values: A small adjusted p (below 0.05) means a real difference; the *fraction of patients where a model is better* says how big it is.

Tiny example: Differences +0.05, +0.02, −0.01, +0.04, +0.03: ranks 5, 2, 1, 4, 3, so $W^+ = 14$, $W^- = 1$, better in 4 of 5. Holm on $p = 0.01, 0.02, 0.04$ gives 0.03, 0.04, 0.04.

Two ways to rank models



Aggregate-then-rank: average each model’s scores, then rank the averages. **Rank-then-aggregate:** rank the models on each patient, then average the ranks.

Like you’re 10: is the best runner the one with the best average time, or the one who wins most races?

Tiny example: A scores 0.9, 0.9, 0.3 (mean 0.70); B scores 0.8 three times (mean 0.80). By averages B wins. By races A wins twice: mean rank A = $(1 + 1 + 2)/3 = 1.33$, B = 1.67, so A wins.

Cohort tests: Kruskal–Wallis and Mann–Whitney

$$H = \frac{12}{N(N+1)} \sum_g \frac{R_g^2}{n_g} - 3(N+1)$$

$$\text{effect size } \varepsilon^2 = \frac{H - k + 1}{N - k}$$

Like you’re 10: Line up all patients from worst to best Dice. If one group (for example venous-phase scans) keeps standing near the front, the groups differ. Two groups use the Mann–Whitney test, more groups the Kruskal–Wallis test (R_g is the sum of the places of group g).

Good values: Small adjusted p : the groups differ. The groups are observational, so a difference can come from other differences between them (site, disease), not only the factor named.

Tiny example: Six patients in two groups taking places $\{1, 2, 3\}$ and $\{4, 5, 6\}$: $H = \frac{12}{42} \left(\frac{36}{3} + \frac{225}{3} \right) - 21 = 3.86$.

3 Results

3.1 Organs and vessels

Table 1: MedFormer on the 901 PanTS test CTs. Mean with 95% bootstrap interval, and median, per structure. The pancreas includes the tumour; the lesion row averages over all patients, including tumour-free ones (see Section 3.3).

	Dice ↑		NSD (2 mm) ↑		HD95 (mm) ↓	
	mean [95% CI]	median	mean [95% CI]	median	mean [95% CI]	median
Pancreas	.864 [.854, .874]	.910	.854 [.842, .864]	.911	19.5 [15.3, 24.5]	2.9
Pancreatic lesion	.597 [.566, .626]	1.000	.590 [.558, .619]	1.000	206.5 [188.4, 225.4]	0.0
Liver	.971 [.965, .975]	.981	.929 [.921, .935]	.961	7.8 [5.7, 10.6]	1.9
Spleen	.928 [.915, .939]	.972	.923 [.910, .935]	.988	31.7 [24.9, 39.6]	1.5
Kidney (left)	.889 [.872, .904]	.966	.880 [.863, .895]	.970	45.4 [36.5, 55.0]	1.8
Kidney (right)	.939 [.929, .947]	.969	.928 [.918, .937]	.972	22.6 [16.9, 28.3]	1.7
Stomach	.903 [.893, .913]	.955	.857 [.846, .869]	.925	16.9 [14.0, 20.1]	2.8
Gallbladder	.805 [.788, .820]	.913	.823 [.806, .838]	.935	56.7 [48.5, 66.0]	2.6
Duodenum	.897 [.890, .904]	.925	.910 [.902, .918]	.942	15.6 [12.1, 19.8]	2.5
Colon	.921 [.915, .926]	.942	.899 [.891, .905]	.927	12.3 [10.3, 14.9]	3.3
Adrenal (left)	.837 [.826, .848]	.877	.928 [.916, .940]	.982	54.4 [46.8, 62.6]	1.5
Adrenal (right)	.843 [.834, .851]	.874	.940 [.930, .949]	.981	60.2 [51.4, 68.9]	1.5
Aorta	.903 [.896, .909]	.936	.914 [.906, .920]	.948	15.0 [13.2, 17.2]	2.7
Inferior vena cava	.854 [.845, .864]	.913	.853 [.843, .862]	.901	22.0 [19.2, 25.4]	4.3
Sup. mesenteric artery	.836 [.828, .844]	.863	.947 [.936, .955]	.982	32.4 [26.4, 39.2]	1.5
Veins	.860 [.852, .866]	.890	.922 [.914, .928]	.949	21.4 [17.7, 25.7]	2.5
Common bile duct	.643 [.626, .659]	.730	.768 [.749, .786]	.889	128.4 [117.7, 140.1]	5.8

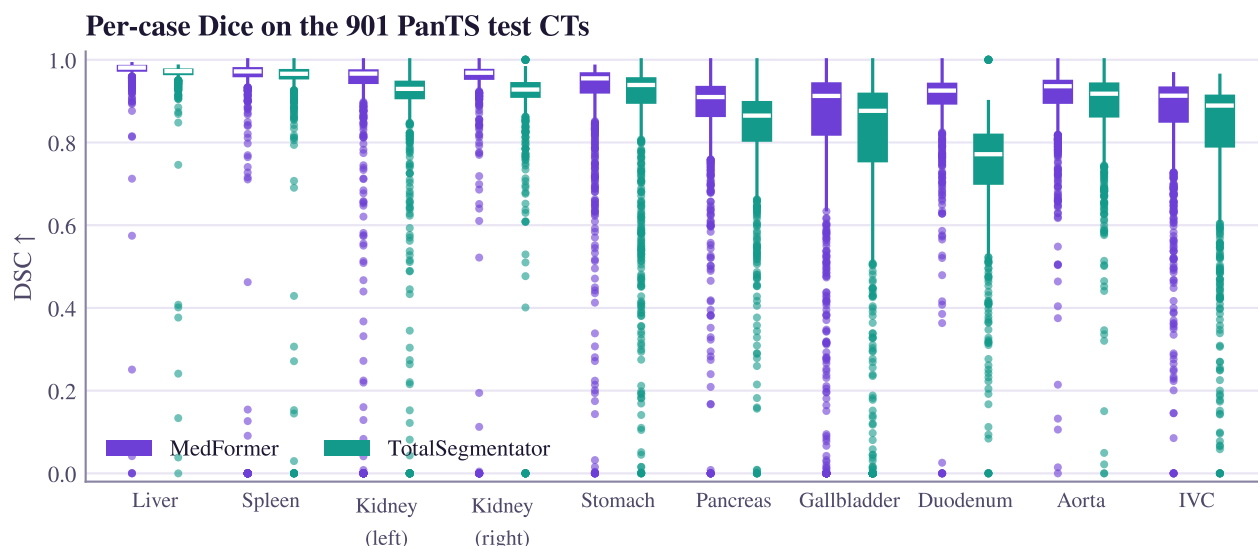


Figure 1: Every patient, not just the average. Per-case Dice of MedFormer and TotalSegmentator on the ten shared organs (box: middle 50% of patients, line: median, dots: outliers).

Medians are higher than means for almost every structure because a minority of patients score far below the rest (Figure 1). The mean HD95 in particular is dominated by a few cases:

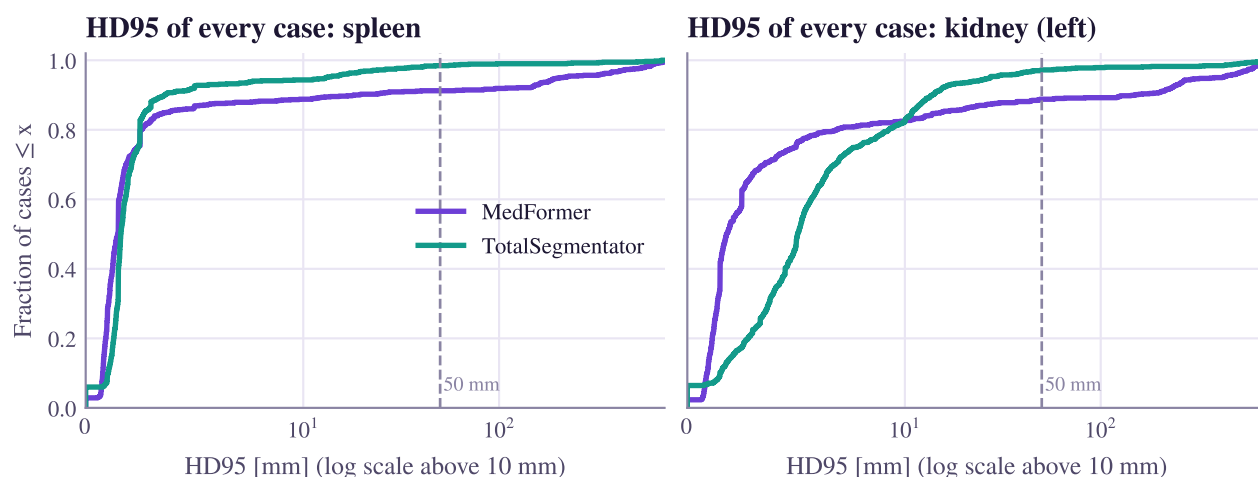


Figure 2: The HD95 failure tail. Fraction of patients with HD95 up to x . Most MedFormer cases are within a few millimetres, but about one in ten is far above 50 mm (dashed).

Where the tail comes from. On the spleen, 79 cases have an HD95 above 50 mm and make up 94% of the mean. In 29 of them the reference has no spleen mask but MedFormer predicts one, which the empty-mask rule scores with the image diagonal (55 test scans have an empty spleen mask; MedFormer predicts a spleen in 29 of them, TotalSegmentator in 1). The other 50 are good segmentations (median Dice .957) with more than 5% of the predicted outline far from the organ (median HD95 180 mm). The left kidney behaves the same way: 101 cases, 93% of the mean, 47 with an empty reference.

3.2 Macro vs. pooled

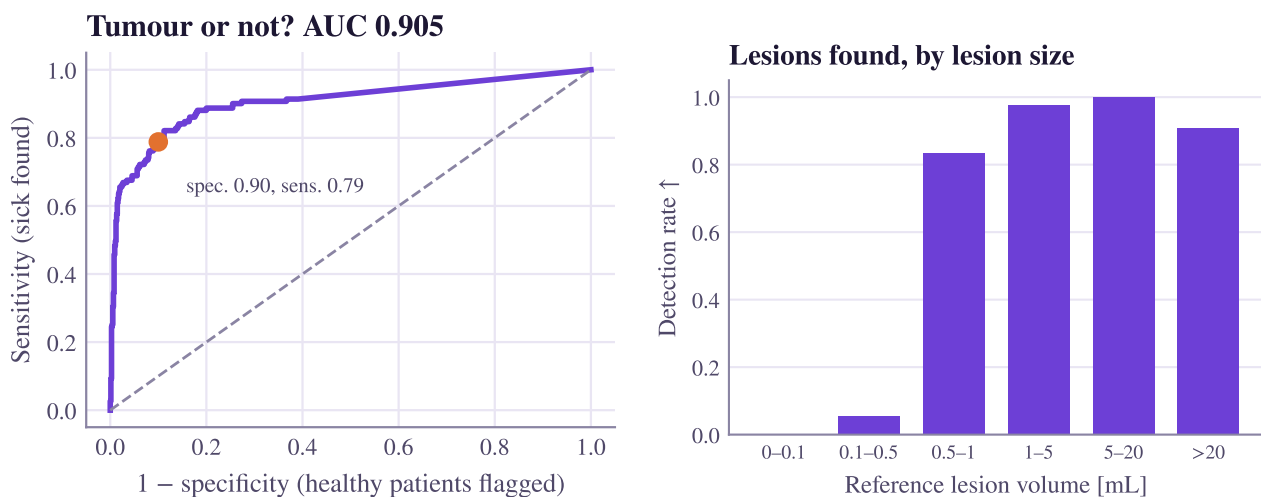
Where the two agree the model is consistent across patients; where pooled is higher (gallbladder .805 macro vs. .879 pooled), a minority of small or unusual structures fails and pulls down the per-patient average.

3.3 Pancreatic tumours

MedFormer finds 131 of the 161 reference tumours, nearly all of those larger than 1 mL (97%), but almost none below 0.5 mL. It also predicts many tumours that are not there, so its lesion precision over the test set is

Table 2: Per-patient (macro) and per-voxel (pooled) Dice. Pooled precision and recall show whether the model tends to over- or under-segment.

Structure	macro Dice	pooled Dice	pooled precision	pooled recall
Pancreas	.864	.855	.817	.897
Liver	.971	.975	.975	.976
Spleen	.928	.965	.962	.967
Kidney (left)	.889	.936	.919	.955
Kidney (right)	.939	.951	.944	.959
Stomach	.903	.928	.916	.940
Gallbladder	.805	.879	.836	.927
Duodenum	.897	.905	.910	.900
Colon	.921	.933	.946	.921
Adrenal (left)	.837	.861	.854	.868
Adrenal (right)	.843	.852	.842	.862
Aorta	.903	.902	.885	.919
Inferior vena cava	.854	.877	.845	.911
Sup. mesenteric artery	.836	.849	.847	.852
Veins	.860	.875	.876	.874
Common bile duct	.643	.692	.614	.793

**Figure 3: Patient-level and lesion-level detection.** Left: ROC curve of the predicted tumour volume as a “tumour or not” score over 151 patients with and 750 without a tumour; the dot is the operating point at specificity ≥ 0.90 (threshold 0.96 mL). Right: fraction of reference tumours found, by tumour size.**Table 3: Tumour detection.** Left: over the whole test set (161 reference tumours; intervals resample patients). Right: reference tumours found, by size.

Test-set lesion metric	value	95% CI	Lesion volume (mL)	lesions	detected
Lesion sensitivity (found / real lesions)	0.814	[0.746, 0.881]	< 0.1	7	0%
Lesion precision (real / predicted lesions)	0.236	[0.199, 0.274]	0.1–0.5	18	6%
Lesion F1	0.356	[0.312, 0.400]	0.5–1	12	83%
False-positive lesions per scan	0.492	[0.436, 0.547]	1–5	44	98%
Pooled lesion Dice (all tumour voxels)	0.552	[0.475, 0.621]	5–20	47	100%
			> 20	33	91%

Table 4: Per-patient tumour metrics (left: means over the 151 patients with a tumour, plus false alarms on the 750 tumour-free patients) and **calibration of the lesion probabilities** (right: voxels within 10 mm of either mask; AP only for patients with a tumour).

Per tumour patient (mean)	value	patients				
Lesion sensitivity	.836	151				
Lesion precision	.821	137				
Lesion F1	.771	151				
Lesion-wise Dice	.486	151				
Panoptic quality	.330	151				
Dice of the tumour voxels	.536	151				
False-positive lesions per scan	0.34	151				
False-positive lesions per scan, tumour-free	0.52	750				

Lesion probability metric	mean	median	patients
Expected calibration error	0.040	0.024	438
Brier score	0.043	0.027	438
Area under PR curve	0.640	0.811	151

low; most of these false tumours are in tumour-free patients (0.52 per scan), and among patients with a tumour the mean per-patient precision is .821 (Table 4). The mean lesion Dice (.597) mixes two groups: on the 750 tumour-free patients a case scores 1 when nothing is predicted (457 patients) and 0 otherwise (mean .609), and on the 151 tumour patients the mean is .536. The pooled lesion Dice of 0.55 describes the tumour voxels themselves.

3.4 Compared with TotalSegmentator

Table 5: Paired comparison on the same 901 patients. Means of MedFormer (MF) and TotalSegmentator (TS) and the fraction of patients in which MedFormer is better. All 30 of 30 Wilcoxon tests are significant after Holm correction (largest adjusted p : $< 10^{-3}$).

Structure	Dice		NSD		HD95 (mm)	
	MF / TS	MF better	MF / TS	MF better	MF / TS	MF better
Pancreas	.864 / .823	91%	.854 / .788	88%	19.5 / 12.2	74%
Liver	.971 / .963	93%	.929 / .898	91%	7.8 / 6.2	81%
Spleen	.928 / .946	79%	.923 / .938	68%	31.7 / 7.6	55%
Kidney (left)	.889 / .890	85%	.880 / .844	82%	45.4 / 16.9	76%
Kidney (right)	.939 / .905	90%	.928 / .853	88%	22.6 / 13.3	80%
Stomach	.903 / .871	90%	.857 / .795	89%	16.9 / 14.7	69%
Gallbladder	.805 / .755	84%	.823 / .755	77%	56.7 / 48.6	69%
Duodenum	.897 / .737	98%	.910 / .669	98%	15.6 / 18.1	92%
Aorta	.903 / .881	80%	.914 / .884	79%	15.0 / 19.9	69%
Inferior vena cava	.854 / .817	88%	.853 / .804	84%	22.0 / 24.3	72%

MedFormer is better in most patients on every organ, including the spleen (79% of patients), where its *mean* Dice is lower than TotalSegmentator’s (.928 vs. .946) because of the failure tail above. The same tail makes MedFormer’s mean HD95 worse on 7 of the ten organs, although it has the lower HD95 in most patients on 10 of the ten. With 901 paired patients every difference is significant, so the fraction of patients and the size of the difference are the informative numbers.

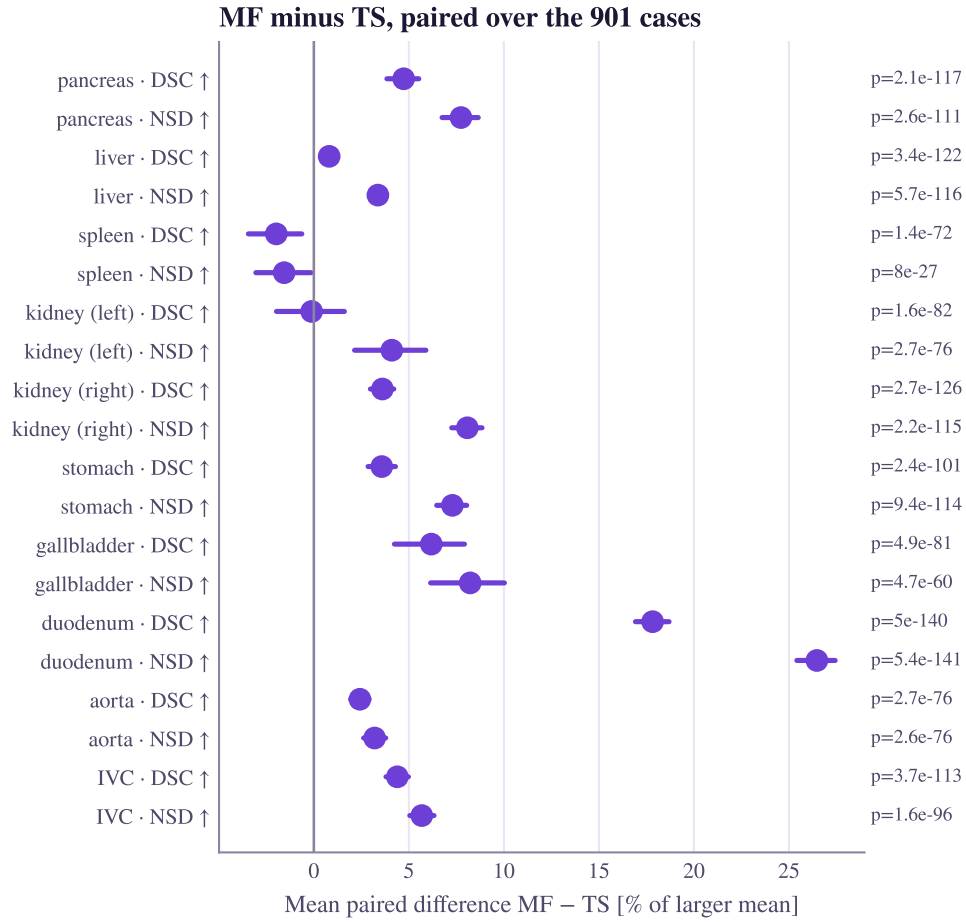


Figure 4: Paired differences with 95% intervals, as a percentage of the larger mean. Right of zero: MedFormer’s mean is higher.

Table 6: Ranking the two models with two schemes, on Dice. Bootstrap τ : median agreement (Kendall’s τ) between the ranking on resampled test sets and the full-data ranking.

Structure	mean Dice MF / TS	winner	mean rank MF / TS	winner	bootstrap τ
Pancreas	.864 / .823	MedFormer	1.09 / 1.91	MedFormer	1.00
Liver	.971 / .963	MedFormer	1.06 / 1.93	MedFormer	1.00
Spleen	.928 / .946	TotalSegmentator	1.18 / 1.79	MedFormer	1.00
Kidney (left)	.889 / .890	TotalSegmentator	1.12 / 1.85	MedFormer	1.00
Aorta	.903 / .881	MedFormer	1.20 / 1.80	MedFormer	1.00

Table 7: MedFormer by patient group. Mean per group of the PanTS metadata (groups under 5 patients and patients without metadata omitted). *Adjusted p*: Kruskal–Wallis or Mann–Whitney test across the groups of a factor, Holm-corrected over all 63 cohort tests. Lesion Dice includes tumour-free patients.

Factor	group	<i>n</i>	pancreas Dice	pancreas NSD	lesion Dice
CT phase	Arterial	162	.838	.832	.612
	Delay	12	.875	.847	.590
	Non-contrast	134	.806	.776	.599
	Venous	593	.884	.878	.593
	<i>adjusted p</i>		$< 10^{-3}$	$< 10^{-3}$	1.000
Scanner	GE	42	.719	.695	.517
	Philips	11	.819	.752	.556
	Siemens	334	.850	.830	.590
	<i>adjusted p</i>		$< 10^{-3}$	0.004	1.000
Country	CH	88	.832	.820	.486
	CN	60	.859	.830	.641
	DE	182	.904	.882	.575
	FR	19	.908	.936	.684
	TR	5	.940	.980	.400
	US	444	.850	.846	.620
	multinational	103	.875	.863	.599
	<i>adjusted p</i>		$< 10^{-3}$	$< 10^{-3}$	1.000
Slice thickness	≤ 1.25 mm	246	.793	.790	.669
	1.25–3 mm	461	.890	.869	.602
	> 3 mm	194	.894	.900	.495
	<i>adjusted p</i>		$< 10^{-3}$	$< 10^{-3}$	0.005
Sex	F	222	.856	.837	.546
	M	269	.843	.830	.643
	<i>adjusted p</i>		1.000	1.000	0.270
Age	< 50	105	.872	.843	.564
	50–65	209	.840	.822	.609
	> 65	165	.863	.859	.599
	<i>adjusted p</i>		1.000	1.000	1.000
Tumour	no tumour	750	.859	.853	.609
	tumour	151	.889	.861	.536
	<i>adjusted p</i>		1.000	1.000	$< 10^{-3}$

3.5 Cohorts

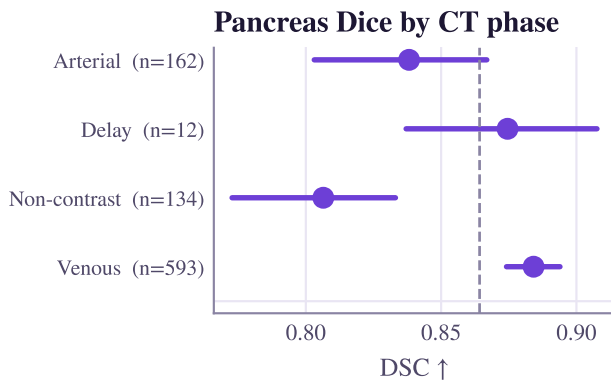


Figure 5: Pancreas Dice by CT phase, mean with 95% interval (dashed: overall mean).

Venous-phase scans are segmented best and non-contrast scans worst ($\text{adjusted } p < 10^{-3}$). Thin-slice scans score *lower* (.793) than thick-slice scans (.894), the opposite of what image resolution alone would predict, which suggests that slice thickness goes together with differences in site and case mix. These groups are observational: they point to where to look, not to causes.

4 Limitations

- **Not a leaderboard number.** PanTS does not publish its evaluation script, NSD tolerance or detection rule; our protocol is documented but may differ from the official one.
- **One code fix.** 25 scans needed a fix of the official padding code (Section 1.2); without it they would have no prediction.
- **Empty reference masks.** Some scans have empty reference masks for organs such as the spleen and kidneys; we do not know whether the organ is absent, outside the scan or not annotated, and the empty-mask rule scores a prediction there as a large error.
- **Tumour-free patients** dominate the mean lesion Dice; read the lesion- and patient-level numbers for tumours.
- **Cohorts are observational**, and some groups are small.
- **Only this R-Super checkpoint.** The report-supervised R-Super models and other training settings were not evaluated.

5 How to reproduce

```
# official inference (GPU; applies the padding fix) and the comparator
sbatch benchmarks/pants/run_medformer.sbatch
sbatch benchmarks/pants/run_totalseg.sbatch
# evaluation and test-set tables (CPU)
sbatch --export=ALL,MODELS=medformer slurm/pants_eval.sbatch
sbatch --export=ALL,MODELS=totalseg slurm/pants_eval.sbatch
segevalkit audit --ref data/raw/PanTS/LabelTe --images data/raw/PanTS/ImageTe \
    --workers 16 --out reports/rsuper_medformer/generated/audit_pants_test.csv
python reports/rsuper_medformer/make_report.py # figures, tables, numbers
tectonic -X compile reports/rsuper_medformer/report.tex
```

Inputs: the per-case results in outputs/eval/pants/medformer/ and .../totalseg/, and the test-set tables in outputs/eval/pants/report/. Code: <https://github.com/aj-das-research/SegEvalKit>.